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On an anti-Ramsey threshold for random graphs[☆]



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ABSTRACT

For graphs G and H , let $G \xrightarrow[p]{rb} H$ denote the property that, for every proper edge-colouring of G (with an arbitrary number of colours) there is a *totally multicoloured*, or *rainbow*, copy of H in G , that is, a copy of H with no two edges of the same colour. We consider the problem of establishing the threshold $p_H^{rb} = p_H^{rb}(n)$ of this property for the binomial random graph $G(n, p)$. More specifically, we give an upper bound for p_H^{rb} and we extend our result to certain locally bounded colourings that generalize proper colourings. Our method is heavily based on a characterization of sparse quasi-randomness given by Chung and Graham (2008).

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1. Introduction

We consider Ramsey and ‘anti-Ramsey’ type properties of the binomial random graph $G(n, p)$. Our terminology and notation are standard (see, e.g., [4,19]).

For graphs G and H , let $G \rightarrow (H)_r$ denote the property that, for every edge-colouring of G with at most r colours, there is a monochromatic copy of H in G . Rödl and Ruciński [30,31] established the threshold function for the Ramsey property $G(n, p) \rightarrow (H)_r$. For simplicity, let us focus on the case in

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which H contains a cycle. The maximum 2-density $m^{(2)}(G)$ of a graph G is given by

$$m^{(2)}(G) = \max \left\{ \frac{e_J - 1}{v_J - 2} : J \subset G, v_J \geq 3 \right\},$$

where e_J and v_J stand for the number of edges and vertices in the graph J , and, for simplicity, we suppose that $v_G \geq 3$. Sometimes we write $e(J)$ for the number of edges in a graph J .

Theorem 1.1 (Rödl and Ruciński [30,31]). *Let H be a graph that contains a cycle. The threshold $p_H = p_H(n)$ for the property $G(n, p) \rightarrow (H)_r$ is given by $p_H(n) = n^{-1/m^{(2)}(H)}$.*

For recent, far reaching generalizations of Theorem 1.1, see Friedgut, Rödl and Schacht [14] and Conlon and Gowers [9]. The anti-Ramsey type property that we are interested in here, expressed by the arrow notation $G \xrightarrow[p]{rb} H$, is the property that, for every proper edge-colouring of G (with an arbitrary number of colours) there is a *totally multicoloured*, or *rainbow*, copy of H in G , that is, a copy of H with no two edges of the same colour.

The relation $G \xrightarrow[p]{rb} H$ was investigated in the case in which G is the random graph $G(n, p)$ and H is the length ℓ cycle C_ℓ by Rödl and Tuza [33]. In fact, by showing that $G(n, p) \xrightarrow[p]{rb} C_\ell$ for fairly small $p = p(n)$, they gave a positive answer to a problem of Spencer (see [11, p. 29]), who asked whether there are graphs of arbitrarily large girth with a rainbow cycle in every proper edge-colouring (for a constructive solution to this problem, see [17]).

Before we proceed, we remark that several authors have considered the problem of finding conditions forcing rainbow or, more generally, richly coloured subgraphs in edge-coloured graphs. The problem we consider here is directly inspired by the work of Rödl and Tuza [33], as is our use of the term ‘anti-Ramsey’, also employed in different contexts. The interested reader will find a wealth of related material in [15,20]. Let us go back to our anti-Ramsey property $G \xrightarrow[p]{rb} H$.

Being an increasing property, $G(n, p) \xrightarrow[p]{rb} H$ admits a threshold function $p_H^{rb} = p_H^{rb}(n)$ for any fixed graph H (see [6]). Here, we focus on *upper bounds* for p_H^{rb} for general graphs H .

Our main result, Theorem 2.1, states that p_H^{rb} is no larger than $n^{-1/m^{(2)}(H)}$, which is the threshold for the Ramsey property $G(n, p) \rightarrow (H)_r$ (see Theorem 1.1). However, as it turns out (see Theorem 2.4), for certain sparse graphs H containing triangles, we have $p_H^{rb} \ll n^{-1/m^{(2)}(H)}$ (we write $f \ll g$ if $f = o(g)$).

As usual, let us say that $G(n, p)$ satisfies a property *almost surely* (a.s.) if the probability that it satisfies the given property tends to 1 as $n \rightarrow \infty$. Let H be a fixed graph. We show that for a suitably large constant C , if $p = p(n) \geq Cn^{-1/m^{(2)}(H)}$, then $G(n, p) \xrightarrow[p]{rb} H$ almost surely. In particular, $p_H^{rb} \leq n^{-1/m^{(2)}(H)}$. In fact, our results can be extended to colourings which are more general than proper colourings (see Section 2).

This paper is organized as follows. Section 2 contains the statement of our main result and in Section 3 we state the main lemmas. In Sections 4 and 5 we prove the main lemmas and the main result, respectively. We finish with some concluding remarks in Section 6.

2. Statement of results

As discussed in the introduction, our main results deal with the threshold $p_H^{rb} = p_H^{rb}(n)$ of the property $G \xrightarrow[p]{rb} H$ for the binomial random graph $G(n, p)$.

Theorem 2.1. *Let H be a graph. Then, for a suitably large constant C , if $p = p(n) \geq Cn^{-1/m^{(2)}(H)}$, then $G(n, p) \xrightarrow[p]{rb} H$ almost surely. In particular, $p_H^{rb} \leq n^{-1/m^{(2)}(H)}$.*

Theorem 2.1 can be extended to certain colourings that generalize proper colourings. A *locally r -bounded edge-colouring* of a graph G is a colouring of its edges such that, for every vertex $v \in V(G)$, no colour appears more than r times on the edges incident to v (in other words, the subgraph of G induced by any single colour has maximum degree bounded from above by r). Note that a proper edge-colouring is just a locally 1-bounded edge-colouring.

For graphs G and H , let $G \xrightarrow[r\text{-loc}]{\text{rb}} H$ denote the property that, for every locally r -bounded edge-colouring of G , there is a rainbow copy of H in G . Let $p_H^{\text{rb}(r)} = p_H^{\text{rb}(r)}(n)$ be the threshold for this property for the binomial random graph $G(n, p)$.

Theorem 2.2 (Generalization of Theorem 2.1). *Let H be a graph and let $r > 0$ be an integer. Then, for a suitably large constant C , if $p = p(n) \geq Cn^{-1/m^{(2)}(H)}$, then $G(n, p) \xrightarrow[r\text{-loc}]{\text{rb}} H$ almost surely. In particular, $p_H^{\text{rb}(r)} \leq n^{-1/m^{(2)}(H)}$.*

A different but related anti-Ramsey property of $G(n, p)$ has recently been investigated by Bohman, Frieze, Pikhurko and Smyth [3]. Let us say that an edge-colouring of G is *r -bounded* if no colour occurs more than r times in total. Let us write $G \xrightarrow[r\text{-bdd}]{\text{rb}} H$ if any r -bounded edge-colouring of G contains a rainbow copy of H and let $p_H^{\text{rb}(r\text{-bdd})}$ be the threshold for this property. Clearly, if $G \xrightarrow[r\text{-loc}]{\text{rb}} H$ then $G \xrightarrow[r\text{-bdd}]{\text{rb}} H$ and hence $p_H^{\text{rb}(r\text{-bdd})} \leq p_H^{\text{rb}(r)}$.

It is proved in [3] that, if H is not acyclic, then $p_H^{\text{rb}(r\text{-bdd})} = n^{-1/m^{(2)}(H)}$, as long as $r = r(H)$ is a large enough constant. In particular, it is proved that this holds for $r \geq 6$ when $H = C_3$, and for $r \geq 5$ when $H = C_\ell$ with $\ell \geq 4$, where C_ℓ denote a cycle of length ℓ . We thus have the following corollary.

Corollary 2.3. *Let H be a non-acyclic fixed graph and let $r = r(H)$ be a large enough positive integer. Then $p_H^{\text{rb}(r)} = n^{-1/m^{(2)}(H)}$. If $H = C_3$, then this holds for any $r \geq 6$, and if $H = C_\ell$ with $\ell \geq 4$, then this holds for any $r \geq 5$.*

We now go back to the case of proper edge-colourings and state a result that shows that p_H^{rb} may sometimes be substantially smaller than $n^{-1/m^{(2)}(H)}$. We first introduce some definitions. Given two graphs H and J such that $m^{(2)}(H) \leq m^{(2)}(J)$, put

$$\beta(H, J) = \min \left\{ \frac{v_{j'} - 2 + 1/m^{(2)}(H)}{e_{j'}} : J' \subset J \text{ with } e_{j'} \geq 1 \right\}.$$

Let $k > 3$. Note that $m^{(2)}(C_k) = 1 + 1/(k-2) < 2 = m^{(2)}(C_3)$. One may check that

$$\beta(C_k, C_3) = \frac{1}{3} \left(1 + \frac{1}{m^{(2)}(C_k)} \right) = \frac{1}{3} \left(1 + \frac{k-2}{k-1} \right).$$

Let F_k be the $(k+1)$ -vertex graph consisting of a k -cycle C_k and a vertex outside this cycle that is incident to two adjacent vertices in the cycle (equivalently, F_k is a $(k+1)$ -cycle with a diagonal that is as short as possible). One may check that

$$-\frac{1}{m^{(2)}(C_k)} < -\beta(C_k, C_3) < -\frac{1}{2} = -\frac{1}{m^{(2)}(F_k)}. \quad (1)$$

In view of (1), Theorem 2.4 shows that the upper bound for $p_{F_k}^{\text{rb}}$ given in Theorem 2.1 may sometimes be improved substantially.

Theorem 2.4. *Suppose $k \geq 4$ and let F_k be the graph defined above. For a suitably large constant C , if $p \geq Cn^{-\beta(C_k, C_3)}$, then $G(n, p) \xrightarrow[p]{\text{rb}} F_k$ holds almost surely. In particular, we have $p_{F_k}^{\text{rb}} \ll n^{-1/m^{(2)}(F_k)}$.*

Theorem 2.4 is a particular case of a result in [24], and will not be proved here.

3. Main lemmas

Let us start with some definitions and notation. Given a graph G and two vertices u and v of G , we denote by $p_t^G(u, v)$ the number of t -paths, that is, paths of length t , between u and v . The number of t -circuits in G , i.e., closed walks of length t , is denoted by $C_t^*(G)$, and $C_t(G)$ denotes the number of t -cycles in G (when counting t -circuits and t -cycles, we count them as ‘labelled’ objects, or, equivalently, we count homomorphisms). The degree of a vertex v is denoted by $\deg_G(v)$ (or simply by $\deg(v)$).

We now introduce a property of graphs that will play an important rôle.

Property 3.1 ($\mathcal{Q}(t, p, \delta, C)$). Suppose $t \geq 2$, $\delta \in (0, 1)$ and $C > 0$ are fixed constants and suppose $0 < p = p(n) \leq 1$ is such that $np \rightarrow \infty$ as $n \rightarrow \infty$. An n -vertex graph G satisfies $\mathcal{Q}(t, p, \delta, C)$ if the following five properties hold.

- (i) $p_j^G(u, v) \leq Cn^{j-1}p^j$ for $j \in \{t, t+1, \dots, 2t+1\}$ and all u and $v \in V(G)$.
- (ii) $|C_{2t+2}^*(G) - (np)^{2t+2}| < \delta(np)^{2t+2}$.
- (iii) $|\deg(v) - np| < \delta np$ for all $v \in V(G)$.
- (iv) $|e(G) - n^2p/2| < \delta n^2p/2$.
- (v) $C_{2t+2}^*(G) \leq C_{2t+2}(G) + \delta(np)^{2t+2}$.

Even though (iv) follows from (iii), we state (iv) explicitly for future reference.

Remark 3.2. If a graph G satisfies $\mathcal{Q}(t, p, \delta, C)$, then $p \geq C^{-1/t}n^{-1+1/t}$ for all large enough n . In fact, since $np \rightarrow \infty$ as $n \rightarrow \infty$ and (iii) holds, we have that $\deg(v) > (1 - \delta)np > t$ for each vertex $v \in V(G)$ for all sufficiently large n . Therefore, G contains a path of length t and hence $p_t^G(u, v) \geq 1$ for some u and $v \in V(G)$. But if $p < C^{-1/t}n^{-1+1/t}$, then $Cn^{t-1}p^t < 1$ and (i) cannot hold.

We now state a lemma that tells us that $G(n, p)$ a.s. has $\mathcal{Q}(t, p, \delta, C)$ for any given $t \geq 2$ and $\delta > 0$ as long as p is a little larger than the lower bound given in Remark 3.2 and C is large enough.

Lemma 3.3. For all $t \geq 2$, there exists $C > 0$ such that if $p \gg (\log n)^2 n^{-1+1/t}$ and $\delta > 0$ is fixed, then $G(n, p)$ satisfies $\mathcal{Q}(t, p, \delta, C)$ almost surely.

We now recall a property of graphs that captures the notion of uniform edge distribution.

Property 3.4 ($\text{DISC}(\varepsilon)$). A graph G satisfies $\text{DISC}(\varepsilon)$ if for all subsets U and $W \subset V(G)$ we have

$$\left| e_G(U, W) - \frac{\text{vol}(U)\text{vol}(W)}{\text{vol}(V(G))} \right| \leq \varepsilon \text{vol}(V(G)),$$

where $\text{vol}(X) = \sum_{x \in X} \deg_G(x)$ for any $X \subset V(G)$. For simplicity, we will denote $\text{vol}(V(G))$ simply by $\text{vol}(G)$.

To state our second main lemma, we need to introduce some further definitions. Let G be an n -vertex graph, let L be a positive integer and consider an edge-colouring $c: E(G) \rightarrow [q]$ of G , where $[q] = \{1, \dots, q\}$. Let $R(L, q)$ be a random subset of $[q]$ such that each colour $i \in [q]$ is present in $R(L, q)$ with probability $1/L$, independently of all the other colours. We denote by $F(G, L, q, c)$ the random set of edges of G whose colours are present in $R(L, q)$. If E' is a set of edges of G , we denote by $G[E']$ the spanning subgraph of G with edge set E' .

Property 3.5 ($\mathcal{P}(L, \varepsilon)$). An n -vertex graph G satisfies $\mathcal{P}(L, \varepsilon)$ if, for any proper edge-colouring $c: E(G) \rightarrow [q]$ of G , the graph $G[F(G, L, q, c)]$ satisfies $\text{DISC}(\varepsilon)$ with probability at least $1 - \varepsilon$.

Together with Lemma 3.3, the next lemma is in the core of the proof of Theorem 2.1.

Lemma 3.6. Let constants $L \geq 1$, $C > 0$, $t \geq 2$ and $\varepsilon \in (0, 1)$ be given. Then there exists $\delta_0 > 0$ such that if an n -vertex graph G satisfies $\mathcal{Q}(t, p, \delta_0, C)$ for some $0 \leq p = p(n) \leq 1$ such that $np \rightarrow \infty$ as $n \rightarrow \infty$, then G satisfies $\mathcal{P}(L, \varepsilon)$ as long as n is large enough.

Our third and last main lemma concerns certain ‘ ε -regular systems’ contained in random graphs. We first need to discuss some rather technical ideas going back to [25].

Let G be a graph and consider subsets $U, W \subset V = V(G)$. We write $E(U, W) = E_G(U, W) = \{\{u, w\} \in E(G) : u \in U, w \in W\}$ for the set of edges of G that have one end-vertex in U and the other end-vertex in W . We set $e(U, W) = e_G(U, W) = |E(U, W)|$. The concepts that follow are related to a variant of Szemerédi’s regularity lemma [35] suitable to cope with sparse graphs (see [23,26]).

Definition 3.7 (*p-Density*). Let H be a graph. Let U and W be non-empty disjoint subsets of $V(H)$ and let $0 < p \leq 1$ be given. The *p-density* of (U, W) in H is given by

$$d_{H,p}(U, W) = \frac{e_H(U, W)}{p|U||W|}.$$

Definition 3.8 (*Regular Pairs*). Let $H = (V, E)$ be a graph and let $\varepsilon > 0$. If U and W are non-empty disjoint subsets of V , then the pair (U, W) is (ε, H, p) -regular if, for all subsets $U' \subset U$ and $W' \subset W$ with $|U'| \geq \varepsilon|U|$ and $|W'| \geq \varepsilon|W|$, we have

$$|d_{H,p}(U', W') - d_{H,p}(U, W)| \leq \varepsilon.$$

If (U, W) is as above and $H = (U, W; E)$ is a bipartite graph, then H is said to be (ε, p) -regular. Furthermore, in the case in which $p = e(U, W)/|U||W|$, we simply say that H is (ε) -regular.

Given a graph H , a real number $\varepsilon > 0$, and positive integers n and m , we define $\mathcal{G}(H, n, m, \varepsilon)$ as the class of graphs with vertex set $V = \bigcup_{x \in V(H)} V_x$, where the V_x have cardinality n each and are pairwise disjoint, and with edge set of the form $E = \bigcup_{\{x,y\} \in E(H)} E_{xy}$, with E_{xy} the edge set of an (ε) -regular graph with exactly m edges between V_x and V_y for all $\{x, y\} \in E(H)$. We denote by $\mathcal{F}(H, n, m, \varepsilon)$ the set of graphs in $\mathcal{G}(H, n, m, \varepsilon)$ that do not contain transversal copies of H (copies of H with exactly one vertex in each V_x).

The next lemma, which allow us to embed graphs in ‘ ε -regular systems’ contained in random graphs, is crucial in our argument.

Lemma 3.9 (*Embedding Lemma (Random Graphs Form)*). Let H be a graph. For any $0 < \alpha < 1$, there exists $\varepsilon > 0$ such that, for all $\mu > 0$, there exists $C > 0$ such that, for all $p \geq Cn^{-1/m^{(2)}(H)}$, a.s. $G(n, p)$ does not contain graphs in $\mathcal{F}(H, \tilde{n}, \alpha\tilde{n}^2p, \varepsilon)$ for any $\tilde{n} \geq \mu n$.

Lemma 3.9, which has by now been proved in a few forms by several authors [2,10,34], belongs to a rich line of research, often useful for tackling embedding problems in subgraphs of random graphs. The interested readers are referred to [32].

4. Proofs of Lemmas 3.3 and 3.6

4.1. Preliminaries

We first introduce two graph properties that will be important in what follows.

Property 4.1 (*CYCLE(t, ε, p)*). An n -vertex graph G satisfies *CYCLE(t, ε, p)* when

$$|C_t(G) - (np)^t| < \varepsilon(np)^t.$$

Property 4.2 (*CIRCUIT(t, ε)*). A graph G satisfies *CIRCUIT(t, ε)* if

$$\left| 1 - \sum_{C_t^*} \prod_{v \in V(C_t^*)} \frac{1}{\deg_G(v)} \right| < \varepsilon,$$

where the sum is over all t -circuits C_t^* in G .

We now state some auxiliary lemmas.

Lemma 4.3 (Chung and Graham [8]). For all $t \geq 2$ and $\varepsilon > 0$, if a graph G satisfies $\text{CIRCUIT}(2t, \varepsilon)$, then G satisfies $\text{DISC}(\varepsilon^{1/2t})$.

Lemma 4.4 (McDiarmid's Inequality [29]). Let $X_1, \dots, X_M$ be independent random variables with X_k taking values on a finite set A_k for each k . Suppose that the function $f: \prod_{1 \leq i \leq M} A_i \rightarrow \mathbb{R}$ satisfies $|f(x) - f(x')| \leq c_k$ whenever the vectors x and x' differ only on the k th coordinate. If Y is the random variable $Y = f(X_1, \dots, X_M)$, then, for any $a > 0$,

$$\mathbb{P}(|Y - \mathbb{E}(Y)| \geq a) \leq 2 \exp \left\{ \frac{-2a^2}{\sum_{1 \leq i \leq M} c_i^2} \right\}.$$

Lemma 4.5 (Chernoff's Inequality). If X is a random variable with distribution $\text{Bi}(n, p)$ and $\lambda = np$, then, for every $0 < \delta \leq 3/2$,

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq \delta \mathbb{E}(X)) \leq 2 \exp \left\{ -\frac{\delta^2}{3} \mathbb{E}(X) \right\}.$$

The next lemma is a small step towards the proof of Lemma 3.6.

Lemma 4.6. Let constants $\delta \in (0, 1)$ and $L \geq 1$ be given. Suppose G is an n -vertex graph with $|\deg_G(v) - np| < \delta np$ for every $v \in V(G)$ and let a proper edge-colouring $c: E(G) \rightarrow [q]$ of G be given. Let $G' = G[F(G, L, q, c)]$. Then

$$\text{for every } v \in V(G') = V(G) \text{ we have } \left| \deg_{G'}(v) - \frac{np}{L} \right| < 3\delta \frac{np}{L} \quad (2)$$

with probability greater than

$$1 - 2n \exp \left(-\frac{(1-\delta)\delta^2}{3L} np \right).$$

Thus, with the same probability,

$$\left| e(G') - \frac{n^2 p}{2L} \right| < 3\delta \frac{n^2 p}{2L}. \quad (3)$$

Proof. Let G, L and c be as in the statement of the result. Note that, for every $v \in V(G)$, the r.v. $\deg_{G'}(v)$ has binomial distribution with parameters $\deg_G(v)$ and $1/L$ (because c is proper). Thus, by Chernoff's inequality (Lemma 4.5), we have that, for every $v \in V(G)$, the inequality

$$\left| \deg_{G'}(v) - \frac{\deg_G(v)}{L} \right| < \delta \frac{\deg_G(v)}{L} \quad (4)$$

holds with probability greater than $1 - 2 \exp \left(-(1-\delta)\delta^2 np / 3L \right)$. Now, since $|\deg_G(v) - np| < \delta np$ for every $v \in V(G)$ and (4) holds, we have that inequality (2) holds with probability greater than

$$1 - 2n \exp \left(-\frac{(1-\delta)\delta^2}{3L} np \right).$$

Inequality (3) now follows from $2e(G') = \sum_{v \in V(G)} \deg_{G'}(v)$. $\square$

We now state a lemma bounding the number of paths of a given length, say j , between two given vertices in a random graph $G(n, p)$. This lemma is a direct consequence of a well known result of Kim and Vu [22]. We need some definitions. In what follows, we suppose $V(G(n, p)) = V(K_n)$. Suppose $I \subset E(K_n)$ is a set of edges and let x and y be two vertices of K_n . Let E_I be the expectation of the number of x – y paths of length j in $G(n, p)$ that contain I , conditional on $I \subset E(G(n, p))$. Put $E_i = \max_{|I|=i} E_I$ and consider $E = \max_{i \geq 0} E_i$ and $E' = \max_{i \geq 1} E_i$. Note that $E_0 = \mathbb{E}(p_j^{G(n,p)}(x, y))$.

Lemma 4.7 (Kim–Vu Inequality for Path Counting [22]). *Let $G = G(n, p)$ and suppose $j \geq 2$ and $x, y \in V(G)$. Then*

$$\mathbb{P}(|p_j^G(x, y) - \mathbb{E}(p_j^G(x, y))| > a_j(EE')^{1/2}\lambda^j) \leq 2e^2e^{-\lambda} \binom{n}{2}^{j-1}$$

for every $\lambda > 1$, where $a_j = 8^j(j!)^{1/2}$.

The following result relates the number of circuits and cycles of a given fixed length in a random graph.

Lemma 4.8. *Let $\ell \geq 2$ and $\delta > 0$ be fixed. If $p \gg n^{-1+1/\ell}$, then we almost surely have*

$$C_{2\ell}^*(G(n, p)) \leq (1 + \delta)(np)^{2\ell}. \quad (5)$$

Proof Sketch. Since $np \gg \log n$, we know that the largest eigenvalue $\lambda_1 = \lambda_1(G)$ of $G = G(n, p)$ is $(1 + o(1))np$. Also, the second largest eigenvalue $\lambda = \lambda(G)$ of G in absolute value is $O(\sqrt{np})$ (the reader is referred to [13] for results concerning the eigenvalues of sparse random graphs). Our result follows from the standard identity

$$C_{2\ell}^*(G) = \text{tr}(A^{2\ell}) = \sum_{1 \leq i \leq n} \lambda_i^{2\ell} = \lambda_1^{2\ell} + \sum_{2 \leq i \leq n} \lambda_i^{2\ell}, \quad (6)$$

involving the number of closed walks of length 2ℓ in G , the trace of the adjacency matrix A of G and the eigenvalues $\lambda_i = \lambda_i(G)$ of G . Now, note that $\lambda_1^{2\ell} = (1 + o(1))(np)^{2\ell}$ and, by our hypothesis on p (i.e., $p \gg n^{-1+1/\ell}$), we have that $\sum_{2 \leq i \leq n} \lambda_i^{2\ell} \leq n\lambda^{2\ell} = O(n(np)^\ell) = o((np)^{2\ell})$. $\square$

We shall use the following corollary of Lemma 4.8.

Corollary 4.9. *Let $\ell \geq 2$ and $\delta > 0$ be fixed. If $p \gg n^{-1+1/\ell}$, then we almost surely have*

$$C_{2\ell}^*(G(n, p)) \leq C_{2\ell}(G(n, p)) + \delta(np)^{2\ell}. \quad (7)$$

4.2. Proof of Lemma 3.2

Fix $t \geq 2$ and let $a_j = 8^j(j!)^{1/2}$ for $j \in \{t, \dots, 2t + 1\}$. Set $C = 1 + a_{2t+1}$ and suppose $p \gg (\log n)^2 n^{-1+1/t}$. Let $G = G(n, p)$ and let $\delta > 0$ be given. We want to prove that a.s. G satisfies $\mathcal{Q}(t, p, \delta, C)$. The fact that $G(n, p)$ a.s. satisfies (ii)–(iv) in the definition of $\mathcal{Q}(t, p, \delta, C)$ for the $p = p(n)$ we are considering is standard. Furthermore, we have already dealt with (v) in Corollary 4.9. To prove (i), we shall use the Kim–Vu inequality (Lemma 4.7).

Let x and y be vertices of G and let $j \geq 2$ be fixed. We are interested in the number of j -paths between x and y in G . Let us compute the quantities E_I , E_i ($i \geq 0$), E and E' introduced before the statement of Lemma 4.7. To compute E_i , one may proceed as follows: fix a set of edges I of the complete graph K_n on $V(G(n, p))$ with $|I| = i$. Now compute the expectation E_I of the number of x – y paths of length j that contain I , conditioned on the event $I \subset E(G(n, p))$. Then $E_i = \max E_I$, where the maximum is taken over all possible sets I .

One may check that, for large enough n , we have $E_i \leq n^{j-i-1}p^{j-i}$ for $i < j$ and $E_j = 1$. Recall that $E' = \max_{i \geq 1} E_i$. As $pn \rightarrow \infty$ as $n \rightarrow \infty$, we have $E' \leq \max\{n^{j-2}p^{j-1}, 1\}$ and $E = \max\{E_0, E'\} \leq \max\{n^{j-1}p^j, E'\}$. We shall treat the cases (a) $j = t$ and (b) $t + 1 \leq j \leq 2t + 1$ separately.

Case (a)(1)

Let $j = t$ and suppose $E' = E_1 = (np)^{t-1}/n$. Thus, $\sqrt{EE'} \leq \sqrt{(np)^{t-1}(np)^t}/n$. Put $\lambda = (np)^{1/2t} \gg n^{1/2t^2}$. We have

$$2e^2 e^{-\lambda} \binom{n}{2}^{t-1} \leq 2e^{2-\lambda} n^{2(t-1)}. \quad (8)$$

By the choice of p , when n tends to infinity, the right-hand side of (8) tends to 0 superpolynomially fast (that is, it is of the form $n^{-\omega}$, with $\omega \rightarrow \infty$ as $n \rightarrow \infty$). Thus, by Lemma 4.7, the probability that $|p_t(x, y) - E_0| > a_t \sqrt{EE'} \lambda^t$ holds tends to 0 superpolynomially fast. Therefore, by the union bound and the fact that E_0 and $\sqrt{EE'} \lambda^t$ are no larger than $(np)^t/n$, we deduce that $p_t(x, y) \leq (1 + a_t)(np)^t/n$ holds for every x and y almost surely.

Case (a)(2)

Let $j = t$ and suppose $E' = 1$. Thus, $\sqrt{EE'} \leq \sqrt{(np)^t}/n$. Now we put $\lambda = (n^{t-1}p^t)^{1/2t} \gg \log n$, we have, by the choice of p , that the right-hand side of (8) tends to 0 superpolynomially fast. Thus, by Lemma 4.7, the probability that $|p_t(x, y) - E_0| > a_t \sqrt{EE'} \lambda^t$ holds tends to 0 superpolynomially fast. We again obtain that $p_t(x, y) \leq (1 + a_t)(np)^t/n$ for all x and y almost surely.

Case (b)

Let $t + 1 \leq j \leq 2t + 1$. Now, $E' = (np)^{j-1}/n$. Thus, we obtain that $\sqrt{EE'} = \sqrt{(np)^{j-1}(np)^j}/n = (np)^j/n\sqrt{np}$. Putting $\lambda = (np)^{1/2j} \gg n^{1/2t(2t+1)}$, we again have, by the choice of p , that

$$2e^2 e^{-\lambda} \binom{n}{2}^{j-1} \leq 2e^{2-\lambda} n^{2(j-1)} \quad (9)$$

tends to 0 superpolynomially fast. Therefore $p_j(x, y) \leq (1 + a_j)(np)^j/n$ holds for all x and y almost surely. In particular, we have verified that (i) in the definition of $\mathcal{Q}(t, p, \delta, C)$ holds almost surely, and the proof of Lemma 3.3 is complete.

4.3. Proof of Lemma 3.6

The argument in the proof of Lemma 3.6 hinges on the following lemma.

Lemma 4.10. For all $L \geq 1, C > 0, t \geq 2$ and $\varepsilon \in (0, 1)$, there exists $\delta > 0$ such that if $np \rightarrow \infty$ as $n \rightarrow \infty$ and an n -vertex graph G satisfies $\mathcal{Q}(t, p, \delta, C)$ then, for every proper edge-colouring $c: E(G) \rightarrow [q]$ of G , the graph $G[F(G, L, q, c)]$ satisfies $\text{CYCLE}(2t + 2, \varepsilon, p/L)$ with probability greater than $1 - \varepsilon$.

Overview of the proof of Lemma 4.10. We wish to show that the random spanning subgraph $G' = G[F(G, L, q, c)]$ of G satisfies $\text{CYCLE}(2t + 2, \varepsilon, p/L)$ with probability greater than $1 - \varepsilon$. Roughly speaking, we want to show that the number of $(2t + 2)$ -cycles in G' is close to $(np/L)^{2t+2}$, with high probability.

First, we show that most $(2t + 2)$ -cycles in G are rainbow. To do this, we use property $\mathcal{Q}(t, p, \delta, C)$ and a counting argument, following Rödl and Tuza [33]. Next, we compute the expected number of rainbow cycles of length $2t + 2$ in G that survive in G' and show that, with high probability, this number is concentrated around its mean, using McDiarmid's inequality. We shall thus have a good estimate for the number of $(2t + 2)$ -cycles in G' , which will tell us that G' satisfies $\text{CYCLE}(2t + 2, \varepsilon, p/L)$ with high probability.

Proof of Lemma 4.10. Fix $L \geq 1, C > 0, t \geq 2$ and $0 < \varepsilon < 1$. Put $\delta = \varepsilon/7$ and consider the following constants: $\alpha = 8\delta^2(1 - 2\delta)^2/(1 + \delta)C^2L^{4t+4}$ and $C' = (1 + \delta)^{t+1}t(2t + 2)C$. Furthermore, put $D = \max\{\log(2/\varepsilon)/\alpha, C'L^{2t+2}/\delta\}$. Suppose $0 < p = p(n) < 1$ is such that $np \rightarrow \infty$ as $n \rightarrow \infty$ and let $n_0 > 0$ be such that $np \geq D$ whenever $n \geq n_0$. Let G be an n -vertex graph with $n \geq n_0$ that satisfies $\mathcal{Q}(t, p, \delta, C)$ (note that, by Remark 3.2, we have $np \geq C^{-1/n^{1/t}}$). Fix a proper edge-colouring $c: E(G) \rightarrow [q]$ of G and denote the random spanning subgraph $G[F(G, L, q, c)]$ of G simply by G' .

We denote by $C_j(G)$, $R_j(G)$ and $N_j(G)$, respectively, the number of cycles, rainbow cycles and non-rainbow cycles of length j in G . Furthermore, $R'(G, j)$ and $N'(G, j)$ denote, respectively, the number of rainbow cycles of length j in G that survive in G' and the number of non-rainbow cycles of length j in G that survive in G' .

First of all, note that, by (iv) in the definition of $\mathcal{Q}(t, p, \delta, C)$, we have

$$e(G) < \frac{(1 + \delta)n^2p}{2}. \quad (10)$$

Now, by making use of an argument of Rödl and Tuza [33], we show that $N_{2t+2}(G) \leq C'(np)^{2t+1}$. Let $\tilde{C}$ be a non-rainbow cycle of length $2t + 2$ in G . Let $k = k(\tilde{C})$ be the minimum length of an arc between two edges sharing the same colour in $\tilde{C}$ (here, by *arc* we mean a path in $\tilde{C}$). Note that $k \leq t$. Let us see how many such cycles $\tilde{C}$ we may have in G , that is, how many non-rainbow copies $\tilde{C}'$ of C_{2t+2} with $k(\tilde{C}') = k$ we may have in G . Let e be one of the edges sharing the same colour with another edge e' in the cycle, with e and e' at distance k in $\tilde{C}$. There exist, by (10), at most $(1 + \delta)n^2p/2$ possible choices for e . By (iii) in the definition of $\mathcal{Q}(t, p, \delta, C)$, the number of k -paths starting at a given vertex is at most $(1 + \delta)^k(np)^k$. This bounds the number of arcs between e and e' . Since the colouring c is proper, there is only one choice for the next edge e' (for e and e' have the same colour). Furthermore, the number of paths of length $(2t + 2) - k - 2 = 2t - k \geq t$ between two vertices is, by (i) in the definition of $\mathcal{Q}(t, p, \delta, C)$, at most $C(np)^{2t-k}/n$. Thus, taking into the account all the possible labellings of the cycles, we conclude that

$$N_{2t+2}(G) \leq \sum_{k=1}^t \frac{C(1 + \delta)^{t+1}(n^2p)(np)^k(np)^{2t-k}2(2t + 2)}{2n} = C'(np)^{2t+1}. \quad (11)$$

Since $R_{2t+2}(G) = C_{2t+2}(G) - N_{2t+2}(G)$, by (11) and (ii) of $\mathcal{Q}(t, p, \delta, C)$, we have that

$$\left(1 - \delta - \frac{C'}{np}\right)(np)^{2t+2} < R_{2t+2}(G) < (1 + \delta)(np)^{2t+2}.$$

Since $np \geq D \geq C'/\delta$, we have $C'/np \leq \delta$. Thus

$$|R_{2t+2}(G) - (np)^{2t+2}| < 2\delta(np)^{2t+2}.$$

But the probability that a rainbow $(2t + 2)$ -cycle in G survives in G' is exactly $1/L^{2t+2}$. Thus

$$\left|\mathbb{E}(R'(G, 2t + 2)) - \left(n\frac{p}{L}\right)^{2t+2}\right| < 2\delta\left(n\frac{p}{L}\right)^{2t+2}. \quad (12)$$

In what follows we shall use McDiarmid's inequality to show that

$$|R'(G, 2t + 2) - \mathbb{E}(R'(G, 2t + 2))| < \delta\mathbb{E}(R'(G, 2t + 2)) \quad (13)$$

holds with probability greater than $1 - \varepsilon$.

For $i \in [q]$, denote by e_i the number of edges having colour i in G . For simplicity, let $X = R'(G, 2t + 2)$. Note that X is a function of $X_1, \dots, X_q$, where $X_i = 1$ if G' has edges with colour i and $X_i = 0$, otherwise. By (i) of property $\mathcal{Q}(t, p, \delta, C)$, we know that if we change the value of X_i , then the value of X will be altered by at most $c_i = e_i C(np)^{2t+1}/n$. Thus,

$$\sum_{i=1}^q c_i^2 = \left(\frac{C(np)^{2t+1}}{n}\right)^2 \sum_{i=1}^q e_i^2. \quad (14)$$

By Lemma 4.4 and (14) we conclude that

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq \delta \mathbb{E}(X)) \leq 2 \exp \left\{ \frac{-2\delta^2 \mathbb{E}(X)^2}{(Cn^{2t} p^{2t+1})^2 \sum_{i=1}^q e_i^2} \right\}.$$

The denominator of the fraction in the argument of the exp function above is largest when the sum $\sum_{i=1}^q e_i^2$ is largest, subject to $\sum_{i=1}^q e_i = e_G$ and $0 \leq e_i \leq n/2$ for all i (the latter constraint comes from the fact that c is a proper colouring). It is not hard to see that this occurs precisely when we have $e_i = n/2$ for as many i as possible. Putting $e_i = n/2$ for $1 \leq i \leq q$ (we ignore integrality conditions, as they are unimportant in the calculations), we have, by (10), that $qn/2 = \sum_{i=1}^q e_i = e(G) < (1 + \delta)n^2 p/2$. Thus $\sum_{i=1}^q e_i^2 \leq (1 + \delta)n^3 p/4$. By (12),

$$\mathbb{P}(|X - \mathbb{E}(X)| \geq \delta \mathbb{E}(X)) \leq 2 \exp \left\{ \frac{-2\delta^2 (1 - 2\delta)^2 (np/L)^{4t+4}}{(Cn^{2t} p^{2t+1})^2 (1 + \delta)n^3 p/4} \right\} \leq e^{-\alpha np},$$

where α is as defined before. Since $np \geq D \geq \log(2/\varepsilon)/\alpha$, we conclude that

$$\mathbb{P}(|X - \mathbb{E}(X)| < \delta \mathbb{E}(X)) > 1 - \varepsilon,$$

i.e., we have proved that (13) holds with probability greater than $1 - \varepsilon$.

Putting (12) and (13) together, we obtain that, with probability greater than $1 - \varepsilon$, we have

$$\left| R'(G, 2t + 2) - \left(n \frac{p}{L}\right)^{2t+2} \right| < 5\delta \left(n \frac{p}{L}\right)^{2t+2}. \quad (15)$$

Since $C_{2t+2}(G') = N'(G, 2t + 2) + R'(G, 2t + 2)$ and $N'(G, 2t + 2) \leq N_{2t+2}(G)$, we conclude, using (11) and (15), that

$$(1 - 5\delta) \left(n \frac{p}{L}\right)^{2t+2} \leq C_{2t+2}(G') \leq \left(1 + 5\delta + \frac{C'L^{2t+2}}{np}\right) \left(n \frac{p}{L}\right)^{2t+2}$$

holds with probability at least $1 - \varepsilon$.

Since $np \geq D \geq C'L^{2t+2}/\delta$, we have $C'L^{2t+2}/np \leq \delta$. Then, as $\delta < \varepsilon/6$, the inequality

$$\left| C_{2t+2}(G') - \left(n \frac{p}{L}\right)^{2t+2} \right| < \varepsilon \left(n \frac{p}{L}\right)^{2t+2}$$

holds with probability greater than $1 - \varepsilon$. $\square$

Overview of the proof of Lemma 3.6. We wish to show that if a graph G satisfies $\mathcal{Q}(t, p, \delta, C)$, then $G' = G[F(G, L, q, c)]$ satisfies $\text{DISC}(\varepsilon)$ with high probability for every proper edge-colouring $c: E(G) \rightarrow [q]$ of G . By Lemma 4.10, the number of $(2t + 2)$ -cycles in G' is concentrated around $(np/L)^{2t+2}$ with high probability. But we know, by Lemma 4.3, that if the number of $(2t + 2)$ -circuits in G' is close to $(np/L)^{2t+2}$, then G' satisfies $\text{DISC}(\varepsilon)$. From (v) of $\mathcal{Q}(t, p, \delta, C)$, we know that the number of $(2t + 2)$ -circuits and the number of $(2t + 2)$ -cycles in G are very close to each other and this can be used to show that the same holds for G' . This fact will complete the proof of Lemma 3.6.

Proof of Lemma 3.6. Fix $L \geq 1$, $C > 0$, $t \geq 2$ and $\varepsilon \in (0, 1)$. Set $D = 1 + L^{2t+2}$ and

$$\delta_1 = \min \left\{ \frac{\varepsilon}{2}, \frac{\varepsilon^{2t+2}}{2D} \right\},$$

$$\delta_2 = \min \left\{ 1 - \left(1 - \frac{\varepsilon^{2t+2}}{2(1 + \varepsilon^{2t+2})} \right)^{1/(2t+2)}, \left(1 + \frac{\varepsilon^{2t+2}}{2(1 - \varepsilon^{2t+2})} \right)^{1/(2t+2)} - 1 \right\}$$

and $\delta = \min\{\delta_1, \delta_2/3\}$. Let δ' be given by an application of Lemma 4.10 with parameters L, C, t and δ . Let $\delta_0 = \min\{\delta, \delta'\}$. We shall show that this choice of δ_0 will do. To that end, suppose that

$0 < p = p(n) < 1$ is such that $np \rightarrow \infty$ as $n \rightarrow \infty$ and that an n -vertex graph G satisfies $\mathcal{Q}(t, p, \delta_0, C)$. We have to show that G satisfies $\mathcal{P}(L, \varepsilon)$ as long as n is large enough.

Let n_0 be such that, for all $n \geq n_0$, we have

$$\left(\frac{n}{C}\right)^{1/t} \geq \frac{3L}{(1-\delta)\delta^2} \log \frac{4n}{\varepsilon}$$

and let $n_1 = 4/\varepsilon$. From now on, we suppose $n \geq \max\{n_0, n_1\}$ and proceed to show that $\mathcal{P}(L, \varepsilon)$ holds.

Fix a proper edge-colouring $c: E(G) \rightarrow [q]$ of G . Since the graph G satisfies $\mathcal{Q}(t, p, \delta', C)$, [Lemma 4.10](#) tells us that, with probability greater than $1 - \delta$, the random spanning subgraph $G' = G[F(G, L, q, c)]$ of G satisfies property $\text{CYCLE}(2t + 2, \delta, p/L)$, i.e.,

$$\left| C_{2t+2}(G') - \left(n \frac{p}{L}\right)^{2t+2} \right| < \delta \left(n \frac{p}{L}\right)^{2t+2}. \quad (16)$$

Since G satisfies $\mathcal{Q}(t, p, \delta, C)$, from [Lemma 4.6](#) we see that, with probability greater than $1 - 2n \exp(-np(1-\delta)\delta^2/3L)$,

$$\text{for every } v \in V(G) \text{ we have } \left| \deg_{G'}(v) - \frac{np}{L} \right| < 3\delta \frac{np}{L}. \quad (17)$$

In the remainder of the proof we assume that (16) and (17) hold and prove that G' satisfies $\text{DISC}(t)$. Note that this conclusion will hold with probability greater than

$$1 - \left(\delta + 2n \exp\left(-\frac{(1-\delta)\delta^2}{3L} np\right) \right). \quad (18)$$

Since $n \geq n_0$ and $p \geq C^{-1/t} n^{-1+1/t}$, we have $np \geq 3L \log(4n/\varepsilon)/(1-\delta)\delta^2$. Therefore, recalling that $\delta \leq \varepsilon/2$, we see that the quantity in (18) is greater than $1 - \varepsilon$. Therefore, it does indeed suffice to prove that (16) and (17) imply that G' satisfies $\text{DISC}(t)$.

We shall first show that G' satisfies $\text{CIRCUIT}(2t + 2, \varepsilon^{2t+2})$ and, after that step, we shall deduce that G' satisfies $\text{DISC}(\varepsilon)$.

By (v) of $\mathcal{Q}(t, p, \delta, C)$ and (16),

$$\begin{aligned} C_{2t+2}^*(G') &= C_{2t+2}(G') + (C_{2t+2}^*(G') - C_{2t+2}(G')) \\ &\leq C_{2t+2}(G') + (C_{2t+2}^*(G) - C_{2t+2}(G)) \\ &\leq (1 + \delta) \left(n \frac{p}{L}\right)^{2t+2} + \delta(np)^{2t+2} \\ &= (1 + \delta(1 + L^{2t+2})) \left(n \frac{p}{L}\right)^{2t+2}. \end{aligned}$$

But, clearly,

$$C_{2t+2}^*(G') \geq C_{2t+2}(G') \geq (1 - \delta) \left(n \frac{p}{L}\right)^{2t+2} \geq (1 - \delta(1 + L^{2t+2})) \left(n \frac{p}{L}\right)^{2t+2}.$$

Thus, we have

$$\left| C_{2t+2}^*(G') - \left(n \frac{p}{L}\right)^{2t+2} \right| < \delta D \left(n \frac{p}{L}\right)^{2t+2}$$

and hence

$$\left| \frac{C_{2t+2}^*(G')}{(np/L)^{2t+2}} - 1 \right| < \delta D.$$

Let us now write $\sum'_{C_{2t+2}^*}$ for the sum ranging over all $(2t + 2)$ -circuits C_{2t+2}^* in G' . By (17),

$$\sum'_{C_{2t+2}^*} \prod_{v \in V(C_{2t+2}^*)} \frac{1}{\deg_{G'}(v)} > \frac{1 - \delta D}{(1 + 3\delta)^{2t+2}}$$

and

$$\sum'_{C_{2t+2}^*} \prod_{v \in V(C_{2t+2}^*)} \frac{1}{\deg_{G'}(v)} < \frac{1 + \delta D}{(1 - 3\delta)^{2t+2}}.$$

Since $\delta \leq \delta_1 \leq \varepsilon^{2t+2}/2D$, we obtain that

$$\frac{1 - \varepsilon^{2t+2}/2}{(1 + 3\delta)^{2t+2}} < \sum'_{C_{2t+2}^*} \prod_{v \in V(C_{2t+2}^*)} \frac{1}{\deg_{G'}(v)} < \frac{1 + \varepsilon^{2t+2}/2}{(1 - 3\delta)^{2t+2}}.$$

Since $\delta \leq \delta_2/3$, the choice of δ_2 let us conclude that

$$\left| 1 - \sum'_{C_{2t+2}^*} \prod_{v \in V(C_{2t+2}^*)} \frac{1}{\deg_{G'}(v)} \right| < \varepsilon^{2t+2},$$

that is, G' satisfies $\text{CIRCUIT}(2t + 2, \varepsilon^{2t+2})$. By Lemma 4.3, we conclude that G' satisfies $\text{DISC}(\varepsilon)$. As observed above, this shows that G satisfies $\mathcal{P}(L, \varepsilon)$, as required. $\square$

5. Proof of the main result

5.1. Preliminaries

The following lemmas will be required later. Our first lemma is sometimes referred to as the ‘slicing lemma’; we give the formulation in [16] (see also Lemma 3.4 in [27]).

Lemma 5.1. *For every $0 < \varepsilon < 1/6$, there exists a constant $C = C(\varepsilon) > 0$ such that any (ε) -regular bipartite graph $B = (V_1, V_2; E)$ contains a spanning (2ε) -regular bipartite subgraph $B' = (V_1, V_2; E')$ with exactly $|E'| = m$ edges, for every m with $C|V_1 \cup V_2| \leq m \leq |E|$.*

Our next lemma relates property DISC and regularity. Given a graph G and disjoint sets of vertices U and $W \subset V(G)$, let $G[U, W]$ be the bipartite graph with vertex classes U and W ‘induced’ by G , that is, with edge set $E_G(U, W)$.

Lemma 5.2. *For every ε and $\mu > 0$ there exist ε' and $\delta > 0$ for which the following holds. Suppose an n -vertex graph G satisfies $\text{DISC}(\varepsilon')$ and, for some $0 < p \leq 1$, we have $|\deg(v) - np| < \delta np$ for every $v \in V(G)$. Then, for every pair of disjoint sets U and $W \subset V(G)$ with $|U|$ and $|W| \geq \mu n$, the induced bipartite graph $G[U, W]$ is (ε) -regular.*

5.2. Proof of Theorem 2.1

We first prove the following deterministic lemma, from which we shall deduce Theorem 2.1.

Lemma 5.3. *For every graph H , there exists $\alpha > 0$ such that for all $t \geq 2$, $C > 0$ and $\varepsilon > 0$, there exists $\delta > 0$ for which the following holds. Let G be an n -vertex graph that satisfies $\mathcal{Q}(t, p, \delta, C)$, where p is such that $np \gg \log n$. Furthermore, let $k = v_H$ and suppose if G does not contain copies of graphs in $\mathcal{F}(H, \lfloor n/k \rfloor, \lfloor \alpha \lfloor n/k \rfloor^2 p \rfloor, \varepsilon)$, then $G \xrightarrow[p]{\text{rb}} H$ for any large enough n .*

Overview of the proof. Let a graph G as in the statement of the lemma be given and fix an arbitrary proper colouring of its edges. We wish to show that there is a rainbow copy of H in this colouring. To achieve this, we shall apply the probabilistic method.

We start by fixing a partition $V_0, V_1, \dots, V_k$ ($k = v_H$) of the vertices of G , where all the V_i ($1 \leq i \leq k$) have the same cardinality and $0 \leq |V_0| < k$. Without loss of generality, we suppose $V(H) = [k] = \{1, \dots, k\}$ and we therefore have a natural bijection between $V(H)$ and the set of the V_i ($i \in [k]$).

We now label each colour used in the proper edge-colouring of G uniformly at random, using the edges of H as labels. For each $e \in E(H)$, we denote by F_e the set of edges of G that are coloured with a colour that has label e . Note that the F_e ($e \in E(H)$) form a partition of $E(G)$. Now, for each $e = \{i, j\} \in E(H)$, let G_e be the bipartite graph with vertex classes V_i and V_j and edge set $E_G(V_i, V_j) \cap F_e$. It will suffice to show that the subgraph $G' = \bigcup_{e \in E(H)} G_e$ of G contains a transversal copy of H , that is, there are $v_i \in V_i$ ($1 \leq i \leq k$) such that $i \mapsto v_i$ is a homomorphism of H into $G' \subset G$. A moment's thought shows that this transversal copy of H in G is indeed a rainbow copy.

Since G satisfies $\mathcal{Q}(t, p, \delta, C)$, we shall be able to apply Lemma 3.6 to conclude that, for every $e \in E(H)$, the random spanning subgraph $G[F_e]$ of G satisfies the discrepancy property (Property 3.4) with high probability. Lemma 5.2 will then tell us that G_e is a regular bipartite graph. Using Lemma 5.1 on each G_e , we shall show that G' contains a copy of a graph in

$$\mathcal{G}\left(H, \left\lfloor \frac{n}{k} \right\rfloor, \left\lfloor \alpha \left\lfloor \frac{n}{k} \right\rfloor^2 p \right\rfloor, \varepsilon\right)$$

with positive probability. Since we suppose that G does not contain copies of graphs in

$$\mathcal{F}\left(H, \left\lfloor \frac{n}{k} \right\rfloor, \left\lfloor \alpha \left\lfloor \frac{n}{k} \right\rfloor^2 p \right\rfloor, \varepsilon\right),$$

we conclude that, with positive probability, G' contains a transversal copy of H , as required.

We now spell out this proof in full.

Proof of Lemma 5.3. Let H be a fixed graph and let $k = v_H$ and $L = e_H$. Fix

$$0 < \varepsilon_1 < \frac{1}{4k^2} \left(\frac{1 - 3/4}{1 + 3/4} \right)^2,$$

and let

$$\alpha = \frac{(1 - 3/4)^2}{(1 + 3/4)L} - 4k^2 \varepsilon_1 \frac{1 + 3/4}{L}.$$

Note that, by the choice of ε_1 , we have $\alpha > 0$. Following the quantification in Lemma 5.3, we let constants $t \geq 2$, $C > 0$ and $\varepsilon > 0$ be given and proceed to find a suitable constant $\delta > 0$. Let $\mu = 1/2k$ and let ε_2 and δ_1 be given by an application of Lemma 5.2 with parameters $\varepsilon/3$ and μ . Let $\varepsilon' = \min\{\varepsilon_1, \varepsilon_2, \varepsilon/2L\}$. Now let δ_2 be given by an application of Lemma 3.6 with parameters L, C, t and ε' . Finally, we let $\delta = \min\{1/4, \delta_1, \delta_2\}$.

Let $0 < p = p(n) \leq 1$ be such that $np \gg \log n$ and consider an n -vertex graph G that satisfies $\mathcal{Q}(t, p, \delta, C)$ and does not contain copies of graphs in $\mathcal{F}(H, \lfloor n/k \rfloor, \lfloor \alpha \lfloor n/k \rfloor^2 p \rfloor, \varepsilon)$ as subgraphs. Now let us fix a proper edge-colouring $c: E(G) \rightarrow [q]$ of G . We shall use the probabilistic method to show that G contains a rainbow copy of H .

For each colour i in $[q]$, we randomly choose one of the L edges of H with uniform probability and independently of all the other colours and assign the selected edge as a label to colour i . Now, if the colouring c gave the colour i to an edge f of G and we selected the edge e of H as the label of the colour i , then we shall say that f has label e . We partition the set $E(G)$ into L subsets F_e ($e \in E(H)$), where each F_e consists of all the edges of G that have label e .

Partition $V(G)$ into $k+1$ sets $V_0, V_1, \dots, V_k$ such that $|V_i| = \lfloor n/k \rfloor \geq \mu n$ for $i \in \{1, \dots, k\}$ and $0 \leq |V_0| < k$. Suppose that $V(H) = [k] = \{1, \dots, k\}$. We associate the set V_i to the vertex i of H , for $i \in [k]$.

For each edge $e = \{i, j\} \in E(H)$, consider the bipartite graph G_e whose vertex classes are V_i and V_j and whose edge set is $E_G(V_i, V_j) \cap F_e$. Consider also the (random) subgraph $G' = \bigcup_{e \in E(H)} G_e$ of G . Our aim is to show that G' contains, with positive probability, a transversal copy of H , which must be rainbow (by construction). We shall then conclude that G contains a rainbow copy of H , as required.

By Lemma 3.6, since $np \rightarrow \infty$ as $n \rightarrow \infty$ and G satisfies $\mathcal{Q}(t, p, \delta_2, C)$ for any fixed $e \in E(H)$, the graph $G[F_e]$ satisfies $\text{DISC}(\varepsilon')$ with probability greater than $1 - \varepsilon'$. Thus, by the union bound, with probability greater than $1 - L\varepsilon'$, every $G[F_e]$ ($e \in E(H)$) satisfies $\text{DISC}(\varepsilon')$.

By (iii) of $\mathcal{Q}(t, p, \delta, C)$ and Lemma 4.6, we know that with probability greater than $1 - 2Lne^{-np(1-\delta)^2/3L}$, for each $e \in E(H)$, the graph $G[F_e]$ is such that,

$$\left| \deg_{G[F_e]}(v) - \frac{np}{L} \right| < 3\delta \frac{np}{L} \quad \text{for every } v \in V(G) \quad \text{and} \quad \left| e_{G[F_e]} - \frac{n^2 p}{2L} \right| < 3\delta \frac{n^2 p}{2L}. \quad (19)$$

In the remainder of the proof we shall assume that each $G[F_e]$ satisfies $\text{DISC}(\varepsilon')$ and that the inequalities in (19) hold. Thus our conclusions will fail with probability at most $L\varepsilon' + 2Ln \exp(-np(1-\delta)^2/3L)$.

Recall that, by the definition of property $\text{DISC}(\varepsilon')$ and the fact that $\varepsilon' \leq \varepsilon_1$, each $G[F_e]$ satisfies

$$\left| e_{G_e}(V_i, V_j) - \frac{\text{vol}(V_i)\text{vol}(V_j)}{\text{vol}(G[F_e])} \right| \leq \varepsilon_1 \text{vol}(G[F_e]),$$

where $\text{vol}(V_i)$ and $\text{vol}(V_j)$ are with respect to the graph $G[F_e]$. Thus, by (19),

$$e(G_e) \geq \lfloor n/k \rfloor^2 p \frac{(1-3\delta)^2}{(1+3\delta)L} - \lfloor n/k \rfloor^2 p \frac{4k^2 \varepsilon_1 (1+3\delta)}{L} > \alpha \lfloor n/k \rfloor^2 p,$$

where the last inequality follows by the choice of δ and α .

Let $\tilde{n} = \lfloor n/k \rfloor$. Since, for each $e \in E(H)$, the graph $G[F_e]$ satisfies $\text{DISC}(\varepsilon')$ (hence, it also satisfies $\text{DISC}(\varepsilon_2)$) and (19) holds, Lemma 5.2 tells us that G_e is $(\varepsilon/3)$ -regular. By Lemma 5.1, each G_e has a spanning subgraph G'_e that is $(2\varepsilon/3)$ -regular, and hence (ε) -regular, with exactly $\lfloor \alpha \tilde{n}^2 p \rfloor$ edges. Thus, $G' = \bigcup_{e \in E(H)} G_e$ contains a copy Γ of a graph in $\mathcal{G}(H, \tilde{n}, \lfloor \alpha \tilde{n}^2 p \rfloor, \varepsilon)$. But by the hypothesis that G does not contain copies of graphs in $\mathcal{F}(H, \tilde{n}, \lfloor \alpha \tilde{n}^2 p \rfloor, \varepsilon)$, the graph Γ must contain a transversal copy of H . Therefore, by the choice of ε' and since $np \gg \log n$, we conclude that with probability greater than

$$1 - \left(L\varepsilon' + 2Ln \exp \left(- \frac{(1-\delta)^2}{3L} np \right) \right) > 0$$

the graph G' contains a transversal copy of H . This copy of H must be rainbow (for edges in different G_e s have different colours). Thus, G contains a rainbow copy of H , as required. $\square$

We are now able to prove Theorem 2.1. Roughly speaking, we just apply Lemma 3.3 and invoke Lemma 3.9 to show that $G(n, p)$ a.s. satisfies the conditions that are required in Lemma 5.3.

Proof of Theorem 2.1. Let H be a fixed graph. Assume that $m^{(2)}(H) > 1$ (in fact, $m^{(2)}(H) = 1$ occurs precisely when H is a forest, and it is easy to see that a simple greedy-type argument proves Theorem 2.1 in this case).

Put $k = v_H$, $L = e_H$, $\mu = 1/2k$ and $t = 2 + \lfloor 1/(m^{(2)}(H) - 1) \rfloor$. Invoke Lemma 5.3 for the graph H to obtain $\alpha > 0$. Apply Lemma 3.9 to H , α , and μ , to obtain constants ε_2 and D . Let C be given by an application of Lemma 3.3 with parameter t . We now carry on with our application of Lemma 5.3, feeding into it the constants C , t and ε_2 , to obtain $\delta > 0$.

Let $p \geq Dn^{-1/m^{(2)}(H)}$ and $G = G(n, p)$. By the choice of t made above, we have that $n^{-1/m^{(2)}(H)} \gg (\log n)^2 n^{-1+1/t}$. Thus,

$$p \gg (\log n)^2 n^{-1+1/t}. \quad (20)$$

Since (20) holds, Lemma 3.3 tells us that G satisfies $\mathcal{Q}(t, p, \delta, C)$ almost surely. By Lemma 3.9, since $p \geq Dn^{-1/m^{(2)}(H)}$, our random graph G a.s. does not contain copies of graphs in $\mathcal{F}(H, \tilde{n}, \lfloor \alpha \tilde{n}^2 p \rfloor, \varepsilon_2)$ for any $\tilde{n} \geq n/2k$ (and thus for $\tilde{n} = \lfloor n/k \rfloor$). By Lemma 5.3, a.s. $G(n, p) \xrightarrow[p]{\text{rb}} H$ holds, as required. $\square$

5.3. Proof of Theorem 2.2

We give a brief sketch of the proof of the generalization of Theorem 2.1 to locally r -bounded colourings.

Sketch of the proof. The proof of Theorem 2.2 is essentially the same as the proof of Theorem 2.1. Besides replacing the condition that the colouring c should be proper by the condition that c should be locally r -bounded in every definition, lemma and theorem, only the following few simple modifications suffice:

- (1) **Proof of Lemma 4.6.** To show the concentration of the degrees of the vertices in the subgraph G' of G we may use, say, Lemma 4.4 (McDiarmid's inequality) instead of Chernoff's inequality.
- (2) **Proof of Lemma 4.10.** Earlier we showed that the number of non-rainbow cycles in G of length $2t + 2$ is given by $N_{2t+2}(G) \leq C'(np)^{2t+1}$. Here, a similar argument to the one used in the proof of Lemma 4.10 gives $N_{2t+2}(G) \leq rC'(np)^{2t+1}$.

Earlier we used that the number e_i of edges having the colour i in a certain subgraph G' of $G(n, p)$ is at most $n/2$ if the colouring is proper. When the colouring is locally r -bounded, we have $e_i \leq rn/2$. Thus, it suffices to multiply the denominator of α by r . $\square$

6. Concluding remarks

We have investigated the threshold function for a certain anti-Ramsey type property for the binomial random graph $G(n, p)$. Standard methods may be used to transfer these results to $G(n, M)$, the random graph on n labelled vertices and M edges, with all such $\binom{\binom{n}{2}}{M}$ graphs equiprobable (see, e.g., [19, Section 1.4]). It would be natural to carry out a similar study for other models of random graphs, such as random graphs with a given degree sequence or appropriately defined W -random graphs (see, e.g., [28, Section 3.1.2] and [5]). Furthermore, we believe that determining the threshold function $p_H^{\text{rb}} = p_H^{\text{rb}}(n)$ remains an interesting open problem.

A variant of the classical Turán extremal function $\text{ex}(n, H)$ has been introduced in [21]. This variant, denoted $\text{ex}^*(n, H)$, is the maximal number of edges in a properly edge-coloured graph on n vertices with no rainbow copy of H . Among other results, it is proved in [21] that, for all non-bipartite graphs H , we have $\text{ex}^*(n, H) = (1 + o(1))\text{ex}(n, H)$. The methods in this paper, together with a standard application of the regularity method, allows one to prove a generalization of this asymptotic identity to subgraphs of random graphs, in the vein of [32, Section 2].

Alon, Jiang, Miller and Pritikin [1] investigated the function $R^*(G, H)$, defined as the smallest n for which every colouring of the edges of the complete graph K_n of order n yields a monochromatic copy of G or a rainbow copy of H (see also [7]). Note that if G is the star with $r + 1$ edges $K_{1,r+1}$, then $R^*(G, H)$ is the smallest n for which $K_n \xrightarrow[r\text{-loc}]{\text{rb}} H$. Here, we investigated the threshold for $G(n, p) \xrightarrow[r\text{-loc}]{\text{rb}} H$. At first sight, it would be natural to generalize this study to the case of pairs of arbitrary graphs (G, H) , given the study of the pairs $(K_{1,r+1}, H)$ presented here. However, it is not very hard to see that the Canonical Ramsey Theorem of Erdős and Rado [12] implies that $R^*(G, H)$ exists if, and only if, either G is a star or H is a forest (see e.g. [18]). This discussion therefore suggests the study of the case in which G is arbitrary and H is a forest. We shall come back to this elsewhere.

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